Modulus Functions & the Factor/Remainder Theorem

ANSWER KEY

Section 1: Solving Modulus Equations

1 Solve the equation $|2x - 5| = x + 1$.

a. $x = 6$ **and** $x = \frac{4}{3}$ ✓

b. $x = 6$ only

c. $x = \frac{4}{3}$ only

d. $x = -6$ and $x = -\frac{4}{3}$

Case 1: $2x - 5 = x + 1 \Rightarrow x = 6$. Check: $|2(6) - 5| = 7 = 6 + 1$. Valid. Case 2: $2x - 5 = -(x + 1) = -x - 1 \Rightarrow 3x = 4 \Rightarrow x = 4/3$. Check: $|8/3 - 5| = 7/3 = 4/3 + 1$. Both are valid.

2 Solve $|3x + 2| = |x - 4|$.

a. $x = \frac{1}{2}$ **and** $x = -3$ ✓

b. $x = 2$ and $x = -3$

c. $x = -\frac{1}{2}$ and $x = 3$

d. $x = 1$ and $x = -4$

Square both sides or branch: $3x + 2 = \pm(x - 4)$. Branch 1: $3x + 2 = x - 4 \Rightarrow 2x = -6 \Rightarrow x = -3$. Branch 2: $3x + 2 = -x + 4 \Rightarrow 4x = 2 \Rightarrow x = 1/2$.

3 Why can $|x - 7| = -3$ have no real solutions?

a. **The absolute value of any real expression is non-negative ($|u| \geq 0$), so it cannot equal a negative number** ✓

b. Because x must be greater than 7

c. Because -3 is not a prime number

d. The equation has complex solutions only

By definition, $|u| \geq 0$ for all $u \in \mathbb{R}$. A non-negative quantity can never equal -3.

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Section 2: Modulus Inequalities

4 Solve the inequality $|2x - 3| < 7$.

a. $-2 < x < 5$ ✓

b. $x < 5$

c. $x < -2$ or $x > 5$

d. $-5 < x < 5$

$|2x - 3| < 7 \Leftrightarrow -7 < 2x - 3 < 7$. Add 3: $-4 < 2x < 10$. Divide by 2: $-2 < x < 5$.

5 Solve $|3x + 1| \geq 10$.

a. $x \leq -\frac{11}{3}$ **or** $x \geq 3$ ✓

b. $-\frac{11}{3} \leq x \leq 3$

c. $x \geq 3$

d. $x \leq -3$ or $x \geq \frac{11}{3}$

$|u| \geq a \Leftrightarrow u \leq -a$ or $u \geq a$. Branch 1: $3x + 1 \leq -10 \Rightarrow 3x \leq -11 \Rightarrow x \leq -11/3$. Branch 2: $3x + 1 \geq 10 \Rightarrow 3x \geq 9 \Rightarrow x \geq 3$.

6 Which interval notation represents the solution to $|x - a| \leq \varepsilon$ where $\varepsilon > 0$?

a. $[a - \varepsilon, a + \varepsilon]$ ✓

b. $(a - \varepsilon, a + \varepsilon)$

c. $[-\varepsilon, \varepsilon]$

d. $[a, a + \varepsilon]$

$|x - a| \leq \varepsilon \Leftrightarrow -\varepsilon \leq x - a \leq \varepsilon \Leftrightarrow a - \varepsilon \leq x \leq a + \varepsilon$, giving $[a - \varepsilon, a + \varepsilon]$.

Section 3: Polynomial Remainder & Factor Theorems

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7 When $P(x) = 2x^3 - 3x^2 + ax - 5$ is divided by $(x - 2)$, the remainder is 11. Find a.

a. 6 ✓

b. 4

c. -6

d. 8

By the Remainder Theorem, $P(2) = 11$. $P(2) = 2(8) - 3(4) + 2a - 5 = 16 - 12 + 2a - 5 = 2a - 1$. Setting $2a - 1 = 11 \Rightarrow 2a = 12 \Rightarrow a = 6$.

8 If $(x + 3)$ is a factor of $Q(x) = x^3 + kx^2 - 7x + 6$, find k.

a. 2 ✓

b. -2

c. 3

d. -1

By the Factor Theorem, $Q(-3) = 0$. $Q(-3) = (-3)^3 + k(-3)^2 - 7(-3) + 6 = -27 + 9k + 21 + 6 = 9k$. Setting $9k = 0 \Rightarrow k = 0$.

9 A polynomial $f(x)$ leaves remainder 3 when divided by $(x - 1)$ and remainder 7 when divided by $(x - 3)$. What is the linear remainder when $f(x)$ is divided by $(x - 1)(x - 3)$?

a. $2x + 1$ ✓

b. $2x - 1$

c. $x + 2$

d. $4x - 1$

Let $R(x) = ax + b$. $R(1) = a + b = 3$ and $R(3) = 3a + b = 7$. Subtracting: $2a = 4 \Rightarrow a = 2$. Then $b = 3 - 2 = 1$. The remainder is $2x + 1$.

Section 4: Transformations of Modulus Graphs

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10 How is the graph of $y = |f(x)|$ obtained from the graph of $y = f(x)$?

a. **Any part of the graph lying below the x-axis is reflected vertically across the x-axis into the upper half-plane ✓**

b. The left half of the graph is discarded and replaced by reflection of the right half

c. The graph is shifted 1 unit upward

d. The graph is reflected across the y-axis

Taking $|f(x)|$ preserves positive y-values and takes the opposite of negative y-values, reflecting sections below the x-axis upward across the x-axis.

11 How is the graph of $y = f(|x|)$ obtained from $y = f(x)$?

a. **Discard the portion for $x < 0$, and reflect the portion for $x \geq 0$ across the y-axis to form an even function ✓**

b. Reflect the entire graph across the x-axis

c. Reflect the graph across $y = x$

d. Shift the graph to the right by $|x|$

Because $|-x| = |x|$, $f(|x|)$ is an even function: the right half ($x \geq 0$) is reflected across the y-axis, discarding the original $x < 0$ part.

12 What is the vertex (sharp turn) of the function $g(x) = 3|x - 4| - 5$?

a. $(4, -5)$ ✓

b. $(-4, -5)$

c. $(4, 5)$

d. $(0, -5)$

The absolute value term $|x - 4|$ reaches its minimum of 0 at $x = 4$. At $x = 4$, $g(4) = 0 - 5 = -5$, giving vertex $(4, -5)$.

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